

CMP2204 Introduction to Computer Networks

Spring 2026, Term Project

Goal: The goal is to apply the concepts learned in class, through programming and hands-on practice. At the end of this project, you will have a better understanding of how a networked application operates and what are the technologies behind it.

Task: Design and implement a peer-to-peer file sharing application. The shared design document (Functional Specification) specifies the necessary protocols that you need to implement. Please follow the design doc closely (in fact, verbatim) in your implementation.

Requirements: The application should;

1. Have 4 processes: `Chunk_Announcer`, `Chunk_Discovery`, `Chunk_Downloader`, `Chunk_Uploader`. These processes should work as outlined in their respective specifications.
2. Successfully detect the available content in the peers in the Local Area Network.
3. End-user interactions: Correctly display detected chunks in the network, display messages when each chunk is downloaded (or cannot be downloaded) from a user, allow the end user to choose from among: displaying users, exchanging numbers to generate secret keys, and requesting chunks to be downloaded in encrypted/unencrypted fashion.
4. Successfully download a content's chunks from other peers in the Local Area Network.
5. Successfully encrypt/decrypt shared chunks when end user prefers security.
6. Output a download/upload log, containing timestamps, names (content name with chunk index) of all downloaded files.

Important Notes: There will be two demo sessions, one on Week 14 and one on Week 15. Those who present in Week 14 will receive 8% bonus. Deadline for the project is 23:59 on **Sunday, June 7** if you will present on Week 14, and on **Sunday, June 14** if you will present on Week 15. (Late submissions will not be accepted for either cases.) Please commit all material under the respective week's assignment folder on MS Teams. Your commits should include:

- All pieces of codes (all python files) that you wrote.
- A README file describing how your program works, and known limitations of your program (so that I run your code correctly).
- A 1-page document describing which platform (Windows/Linux/MacOS) you've used to develop your code, faced challenges, group members' names and division of workload within the group.
- Wireshark capture screenshots of one encrypted and one unencrypted chunk exchange, showing you sending a text containing the names of your team members. Make sure you capture the png header in the unencrypted version.

Please name all your files as [XXX]_[filename] where XXX is **your team name** from the ProjectTeams.xlsx on Teams.

You may work in groups of size **2 or 3**. You should determine a partitioning of responsibilities so that group members can work efficiently in parallel.

Please note that only the students who present their work in the demo session on weeks 14-15 will get a grade out of their term project.